★ DJ ALGORITHM - Contd

→ We are given a black box known as an oracle which implements a function defined by

$$f : \{0,1\}^n \rightarrow \{0,1\}$$

The function takes n-bit binary input & produces either 0 or 1 as output. We are guaranteed that the func f is either constant or balanced. The task is to determine if f is constant or balanced using the oracle least no of times.

① Take 2 qubit system with each of the qubits initialised to 1.

$$|\psi_1\rangle = |11\rangle$$

② Apply Hadamard gate on each of the qubits to produce superposition

$$|\psi_2\rangle = H \otimes H\, |\psi_1\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle - |01\rangle - |10\rangle + |11\rangle\right]$$

$$= \cancel{\frac{1}{\sqrt{2}}\left[|01\rangle + |10\rangle - 2|11\rangle\right]}$$

$$= \frac{1}{2}\left[[|0\rangle - |1\rangle] \otimes [|0\rangle - |1\rangle]\right]$$

③ Apply the function box followed by CNOT on 2nd qubit.

$$|\psi_3\rangle = f \cdot CNOT|\psi_2\rangle$$

bit-wise addition

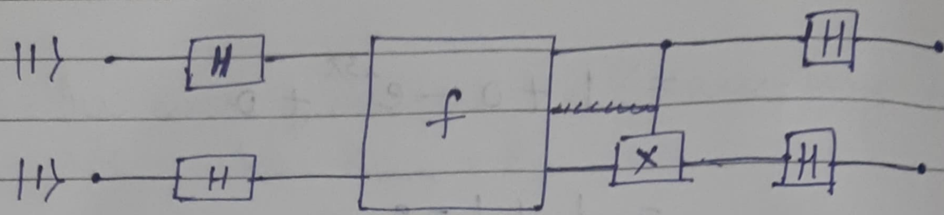
$$|\psi_3\rangle = \frac{1}{2}\left[|0\, f(0_1) \oplus 0\rangle + \ldots\ldots\right]$$

④ Apply H gate on 2nd qubit.

⑤ Measure all

If after measurement, output is 0 then constant mode or else balanced mode.

DJ Algo



$$|\psi\rangle = \frac{1}{2}\left[\,|00\rangle - |01\rangle - |10\rangle + |11\rangle\,\right]$$

$$f\text{-}CX\,|00\rangle = |0\; f(0) \overset{XOR}{\oplus} 0\rangle = |0\, f(0)\rangle$$

$$f\text{-}CX\,|01\rangle = |0\; f(0) \oplus 1\rangle = |0\, \overline{f(0)}\rangle$$

$$f\text{-}CX\,|10\rangle = |1\; f(1) \oplus 0\rangle = |1\, f(1)\rangle$$

$$f\text{-}CX\,|11\rangle = |1\; f(1) \oplus 1\rangle = |1\, \overline{f(1)}\rangle$$

$$f\text{-}CX\,|\psi\rangle = \frac{1}{2}\Big[\,|0\, f(0)\rangle - |0\, \overline{f(0)}\rangle - |1\, f(1)\rangle + |1\, \overline{f(1)}\rangle\,\Big]$$

$$= \frac{1}{2}\Big[\,|0\rangle \otimes \left[\,|f(0)\rangle - |\overline{f(0)}\rangle\,\right] - |1\rangle \otimes \left[\,|f(1)\rangle - |\overline{f(1)}\rangle\,\right]\Big]$$

Case 1: Balanced mode

It depends on input

$f(|0\rangle) = |1\rangle = \overline{f(|0\rangle)}$

$f(|1\rangle) = |0\rangle = \overline{f(|0\rangle)}$

$f(|1\rangle) = |0\rangle = \overline{f(|1\rangle)}$

$f(|0\rangle) = |1\rangle = \overline{f(|1\rangle)}$

$f\text{-}cx\,|\psi\rangle = \frac{1}{2}\left[|0 f(0)\rangle - |1 f(1)\rangle - |0 f(1)\rangle + |1 f(0)\rangle\right]$

$= \frac{1}{2}\left[f(|0\rangle) \otimes [|0\rangle + |1\rangle] - f(1\rangle \otimes [|0\rangle + |1\rangle]\right]$

$|\psi_0'\rangle = \frac{1}{2}\left[\overset{\to q_2}{f(|0\rangle) - f(|1\rangle)}\right] \otimes \overset{\to q_1}{[|0\rangle + |1\rangle]}$

Case 2: Constant mode

It is independent of input.

$f(|0\rangle) = |0\rangle$

$f(|1\rangle) = |0\rangle = f(|0\rangle)$

$f(|1\rangle) = |1\rangle$

$f(|0\rangle) = |1\rangle = f(|1\rangle)$

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$$f\text{-}cx|\psi\rangle = \frac{1}{2}\left[\overset{q_1}{|0\rangle - |1\rangle}\right] \otimes \left[\overset{q_2}{f(|0\rangle) - f(|1\rangle)}\right]$$

$$|\psi'_c\rangle = |-\rangle \otimes \frac{1}{\sqrt{2}}\left[f(|0\rangle) - f(|1\rangle)\right]$$

Applying Hadamard on q_1

$$H|\psi'_b\rangle = |0\rangle \otimes \frac{1}{\sqrt{2}}\left[f(|0\rangle) - f(|1\rangle)\right]$$

$$H|\psi'_c\rangle = |1\rangle \otimes \frac{1}{\sqrt{2}}\left[f(|0\rangle) - f(|1\rangle)\right]$$

$\because$ $\left[H|+\rangle = |0\rangle \quad H|-\rangle = |1\rangle\right]$

Both the function box & CNOT didn't alter qubit 1

So, k

After

no

→ QFT

QFT

$|0\rangle$ x

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* SHOR'S FACTORIZATION ALGO

Step 1: Check if its an even or odd

Step 2: check if it prime or not

Step 3: Pick a random no in range [2, 34].

Step 4: Check for common factors.

Step 5: Find r when $f(x) = a^x \% N = 1$

Step 6: Find $\gcd(a^{r/2} \pm 1, N)$

the gcd fails when r is odd or when $N^x = a^{r/2} \pm 1$

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* SHOR'S ALGORITHM

→ Quantum fourier transforms converts the graph of frequency vs relative amplitude to discrete frequency graph.

→ Consider a sample given as

$$x = [1, 0, -1, 0] \quad \text{(amplitude in \{time\} domain)}$$

$N=4$

Calculate the discrete fourier transform of the given sample & obtain the constant frequency.

$$x \rightarrow X[k]$$

where $k=0$, base
$k=1$, 1^{st} harmonic, so on..

$$X[k] = \sum_{n=0}^{N-1} e^{i2\pi k n/N} x[n]$$

This is called as discrete fourier transforms.

N - no of sample points

$k_{max} = \infty$ mathematically but in practical $k_{max} = N/2$.

if $k=0$, in signal processing, it is DC signal
, in analog frequency based its invalid.

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→ Higher sampling rate, more clarity in audio.

In our question, N (sampling rate) = 4, so $k_{max} = 2$

$$X[k=0] = \sum_{n=0}^{3} x[n] = 0$$

↳ Magnitude in freq domain

$$X[k=1] = \sum_{n=0}^{3} e^{i2\pi n/4} x[n]$$

$$= e^{0} \times 1 + e^{i\pi/2} \times 0 + e^{i\pi} \times -1 + e^{i3\pi/2} \times 0$$

$$= 1 + e^{i\pi} \times -1 =$$

$$e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1$$

$$X[k=1] = 1 + 1 = 2$$

$$X[k=2] = \sum_{n=0}^{3} e^{i\pi n} x[n]$$

$$= 1 + 0 - e^{2\pi i} + 0$$

$$= 1 - 1 = 0$$

$$X[k=3] = \sum_{n=0}^{3} e^{i3\pi n/2} \cdot x[n]$$

$$= 1 + 0 - e^{i3\pi} + 0$$

$$= 1 + 1 = 2_4$$

Q) $x = [1, 0, -1, 0, 1, 0, -1, 0]$

$N = 8$ $\qquad k_{max} = 4$

$k = 0$ $\qquad X[k] = 0$

$k = 1$ $\qquad X[k] = \sum_{n=0}^{7} e^{i\pi n/4} x[n]$

$$= 1 + 0 - e^{\pi i/2} + 0 + e^{\pi i} + 0 - e^{3\pi i/2} + 0$$

$$= 1 + 1 - 1 + 1 = 2_4$$

$k = 2$ $\qquad X[k] = \sum_{n=0}^{7} e^{i\pi n} x[n]$

$$= 1 + 0 - e^{2i} + 0 + e^{4i} + 0 - e^{6i} + 0$$

$$= 1 - 1 + 1 - 1 = 0\,4$$

$k = 3$ $\qquad X[k] = \sum_{n=0}^{7} e^{i3\pi n/2} x[n]$

$$= 1 + 0 - e^{3\pi i} + 0 + e^{i6\pi} + 0 - e^{12i\pi} + 0$$

$$= 1 + 1 + 1 - 1 = 2_4$$

So, $k=2$ is part of our signal

After 4 points, repetition takes place

so $k=2, 6, 10, \dots$ so on are part of our signal.

$$y(t) = \cos(2\pi f k t)$$

→ QFT

normalization factor

$$QFT|\psi_i\rangle = \frac{1}{\sqrt{N}} \sum_{n=1}^{N-1} e^{i2\pi \psi_i n/N} |n\rangle$$

Q) $x = [1, 0, -1, 0]$

ψ_i	$\vert\psi_i\rangle$	n	$\vert n\rangle$
0	$\vert 00\rangle$	0	$\vert 00\rangle$
1	$\vert 01\rangle$	1	$\vert 01\rangle$
2	$\vert 10\rangle$	2	$\vert 10\rangle$
3	$\vert 11\rangle$	3	$\vert 11\rangle$

Represents time domain — represents frequency domain

$$|\psi\rangle = \frac{1}{\sqrt{2}} \sum_{x=0}^{N-1} x[i]\,|\psi_i\rangle =$$

$$= \frac{1}{\sqrt{2}} \left[1|00\rangle + 0|01\rangle - 1|10\rangle + 0|11\rangle \right]$$

$$= \frac{1}{\sqrt{2}} \left[|00\rangle - |10\rangle \right]$$

$\because$ total $P = 1$

So $|\psi\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle - |10\rangle\right]$

$QFT|\psi\rangle = \frac{1}{\sqrt{2}}\left[QFT|00\rangle - QFT|10\rangle\right]$

$$QFT|00\rangle = \frac{1}{2}\Big[e^{i2\pi \times 0 \times 0/4}|00\rangle + e^{i2\pi 0 \times 1/4}|01\rangle + e^{i2\pi \times 0 \times 2/4}|10\rangle + e^{2\pi i \times 0 \times 3/4}|11\rangle\Big]$$

$$= \frac{1}{2}\left[|00\rangle + |01\rangle + |10\rangle + |11\rangle\right]$$

$$QFT|10\rangle = \frac{1}{2}\left[|00\rangle - |01\rangle + |10\rangle - |11\rangle\right]$$

$$QFT|\psi\rangle = \frac{1}{2\sqrt{2}}\left[2|01\rangle + 2|11\rangle\right]$$

$$QFT|\psi\rangle = \frac{1}{\sqrt{2}}\left[|01\rangle + |11\rangle\right]$$

$\lvert\psi_i\rangle$	$P(\lvert\psi_i\rangle)$	k	$x[k]$
$\lvert 00\rangle$	0	0	0
$\lvert 01\rangle$	$\underline{1/2}$	1	$\underline{2}$
$\lvert 10\rangle$	0	2	0
$\lvert 11\rangle$	$\underline{1/2}$	3	$\underline{2}$